Chapter 1
Matrices and Systems of Equations

1.2 Row Echelon Form

Learning outcomes

1. Be able to tell what is row echelon form and reduced row echelon form (能够辨识什么是行阶梯形和行最简行形的矩阵)
2. Be able to obtain the (reduced) Row Echelon Form of a matrix 会将一个矩阵通过行初等变换转化为等价的行阶梯形、行最简形矩阵
3. Be able to determine whether a linear system is consistent according to its row echelon form. 能够通过线性方程组增广矩阵的等价行阶梯形矩阵判断方程组是否有解， 有几个解
4. Be able to obtain the solutions of consistant linear system, 会求解相容线性方程组的解。
5. Be able to solve under-determined linear systems, determine the free variables according to the row echelon form. 能够求解亚定方程组，并根据行阶梯形确定自由变量。

Ch. 1 : nxn linear system. S.T.F

But.

(1). How to solve a linear system that cannot be simplified to S.T.F ?

(2) mxn.

How to find the complete solution set for any mxn LS ?

Example 1

Solving the following system of linear equations

$$x_1 + 2x_2 + 3x_3 + 2x_4 = -1$$
$$-x_1 - 2x_2 - 2x_3 + x_4 = 2$$
$$2x_1 + 4x_2 + 8x_3 + 12x_4 = 4$$

3×4

Example 1

Solving the following system of linear equations

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$$2x_1 + 4x_2 + 8x_3 + 12x_4 = 4$$

Solution

Starting with the augmented matrix, create zeros below the pivot in the first column.

$$\begin{bmatrix} 1 & 2 & 3 & 2 & | & -1 \\ -1 & -2 & -2 & 1 & | & 2 \\ 2 & 4 & 8 & 12 & | & 4 \end{bmatrix} \sim \begin{matrix} \\ R2+R1 \\ R3+(-2)R1 \end{matrix} \begin{bmatrix} 1 & 2 & 3 & 2 & | & -1 \\ 0 & 0 & 1 & 3 & | & 1 \\ 0 & 0 & 2 & 8 & | & 6 \end{bmatrix}$$

At this stage, we create a zero only below the pivot.

$$\sim \begin{matrix} \\ \\ R3+(-2)R2 \end{matrix} \begin{bmatrix} 1 & 2 & 3 & 2 & | & -1 \\ 0 & 0 & 1 & 3 & | & 1 \\ 0 & 0 & 0 & 2 & | & 4 \end{bmatrix} \sim \begin{matrix} \\ \\ \frac{1}{2}R3 \end{matrix} \begin{bmatrix} 1 & 2 & 3 & 2 & | & -1 \\ 0 & 0 & 1 & 3 & | & 1 \\ 0 & 0 & 0 & 1 & | & 2 \end{bmatrix}$$

Echelon form

The corresponding system of equation is

$$x_1 + 2x_2 + 3x_3 + 2x_4 = -1$$
$$x_3 + 3x_4 = 1$$
$$x_4 = 2$$

$x_3 = 1 - 3x_4 = -5$

We get

$$x_3 + 3(2) = 1$$
$$x_3 = -5$$

Substituting $x_4 = 2$ and $x_3 = -5$ into the first equation,

$$x_1 + 2x_2 + 3(-5) + 2(2) = -1$$
$$x_1 + 2x_2 = 10$$
$$x_1 = -2x_2 + 10$$

Let $x_2 = \alpha$ $\forall \alpha \in R$

Let $x_2 = r$. The system has many solutions. The solutions are

$$x_1 = -2r + 10,\ x_2 = r,\ x_3 = -5,\ x_4 = 2$$

α can be any real number

$(-2\alpha + 10, \alpha, -5, 2)$ $\forall \alpha \in R$ is the solution.

x_2 is referred to as free variable; x_1 x_3 x_4 are referred to as lead variables, i.e., the variables correspond to the first nonzero elements in each row of the reduced matrix.

How to determine free variables and leading variables 如何确定自由变量

$$\overset{\frac{1}{2}R3}{\approx}\left[\begin{array}{cccc|c} 1 & 2 & 3 & 2 & -1 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

x_1 x_2 x_3 x_4

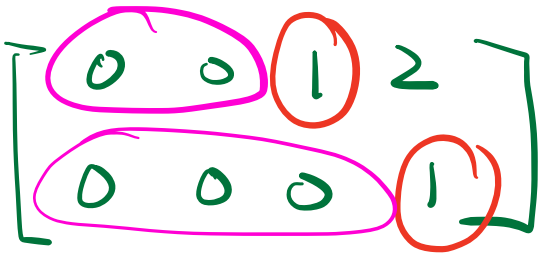
x_1, x_3, x_4 are leading variables.

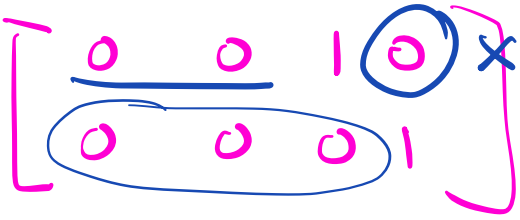
The variable that is not a leading variable is a free variable.

- Unknowns corresponding to columns with lead entries are lead variables
- Unknowns corresponding to columns with no leading entries are free variables. Free variables can be any number. 自由变量的可以是任意值，它对应于没有lead entries 的列。

Express the leading variable in terms of free variables.

Leading 1: The first 1 in a non-zero row in the echelon form.





Leading 0s: The zero entries in a row before the first non-zero entry.

A linear system can have many free variables.

Eg. $\left[\begin{array}{ccccc|c} 0 & 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & 0 & 1 & 0 & 4 \end{array}\right]$ 2×5

x_1 x_2 x_3 x_4 x_5

How to express solutions?

$x_3+2x_4+x_5=3$

$x_4=4$

x_3, x_4 are leading variables.

x_1, x_2, x_5 are free variables.

$x_3=3-8-x_5$

$=-5-x_5$

$=-5-\gamma$

Let $x_1=\alpha \quad \forall \alpha\in\mathbb{R}$

$x_2=\beta \quad \forall \beta\in\mathbb{R}$

$x_5=\gamma, \quad \forall \gamma\in\mathbb{R}$

$(\alpha, \beta, -5-\gamma, 4, \gamma) \quad \forall \alpha,\beta,\gamma\in\mathbb{R}$

Row Echelon Form/行阶梯形矩阵

$$\begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{bmatrix} \rightarrow \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3×4

Definition

A matrix is in **echelon form** if

1. If there are rows whose entries are all zero, they are below the rows having nonzero entries. 都是零的行在最后面
2. The first nonzero element of each row is 1. This element is called a **leading 1**. 每一行中第一个非零元素是1
3. If row k does not consist entirely of zeros, the number of leading zero entries in row $(k+1)$ is greater than the number of leading zero entires in row k. 下面的零比上面的多

zero row : $[0\ 0\ 0\ 0]$

non-zero row : $[0\ 0\ 1\ 0]$

$[1\ 2\ 3\ 4]$

↓

exist some non-zero entries in the row

1. If there is a zero row, the zero row is below the non-zero row.

If there is no zero row, no need to consider Requirement 1.

e.g. ① $\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ ✓ r.e.f

② $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ leading 1 ✓ r.e.f.

Examples for Row Echelon Form

$$\begin{bmatrix} 1 & 0 & 8 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 9 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 1 & 3 & 0 \\ 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

1> ✓

2> ✗

3> ✓

✗

r.e.f

Examples for Row Echelon Form

$$\begin{bmatrix} 1 & 0 & 8 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 9 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 & 3 & 0 \\ 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(×) (×) (✓) (✓)

The process of using row operations to transform a linear system into one whose augmented matrix is in row echelon form is called **Gaussian elimination**

(高斯消元法) → r.e.f

an m×n LS → augmented matrix

Type I、II、III → r.e.f → decide leading & free variables

Note

DID YOU KNOW?

Carl Friedrich Gauss, the great German mathematician, did not discover what is called Gaussian elimination.

The method of Gaussian elimination appears in Chapter 8, *Rectangular Arrays*, of the important Chinese mathematical text *Jiuzhang suanshu* or *The Nine Chapters on the Mathematical Art*. Its use is illustrated in eighteen problems, with two to five equations. The first reference to the book by this title is dated to 179 CE, but parts of it were written as early as approximately 150 BCE. It was commented on by Liu Hui in the 3rd century.

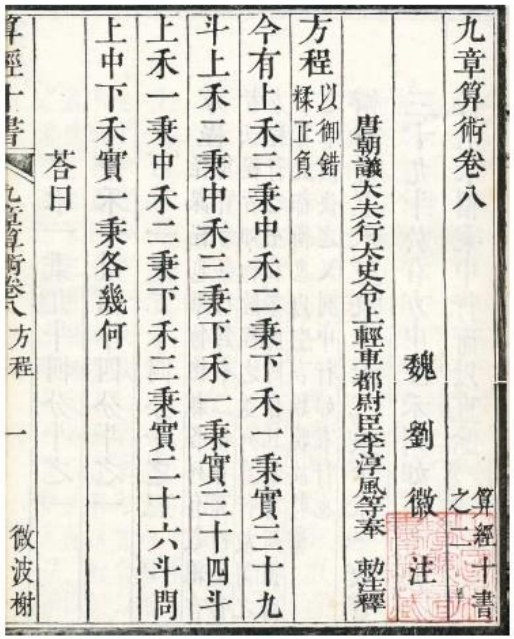
The method in Europe stems from the notes of Isaac Newton. In 1670, he wrote that all the algebra books known to him lacked a lesson for solving simultaneous equations, which Newton then supplied. Cambridge University eventually published the notes as *Arithmetica Universalis* in 1707 long after Newton left academic life. The notes were widely imitated, which made (what is now called) Gaussian elimination a standard lesson in algebra textbooks by the end of the 18th century. Carl Friedrich Gauss in 1810 devised a notation for symmetric elimination that was adopted in the 19th century by professional hand computers to solve the normal equations of least-squares problems. The algorithm that is taught in high school was named for Gauss only in the 1950s as a result of confusion over the history of the subject.

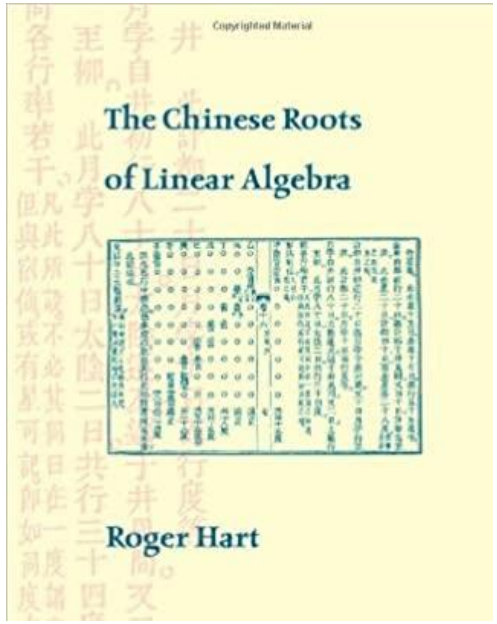
Note

Example of systems of equations



Liu Hui
3rd Century AD





Problem 1, Chapter 8 of Jiu Zhang Suan Shu



$$\begin{cases} 3x+2y+z=39 \\ 2x+3y+z=34 \\ x+2y+3z=26 \end{cases}$$

Gauss Elimination

注意古文的阅读顺序是从上到下，从左到右

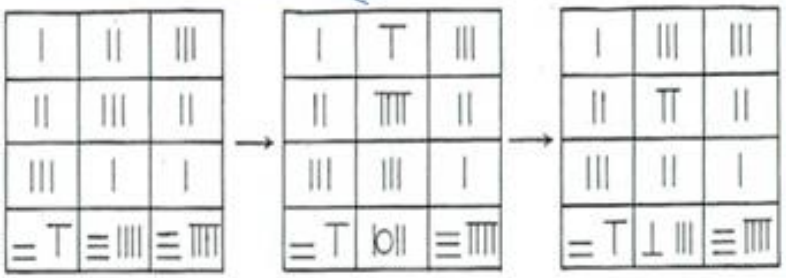


$$\begin{cases} 3x+2y+z=39 \\ 2x+3y+z=34 \\ x+2y+3z=26 \end{cases}$$

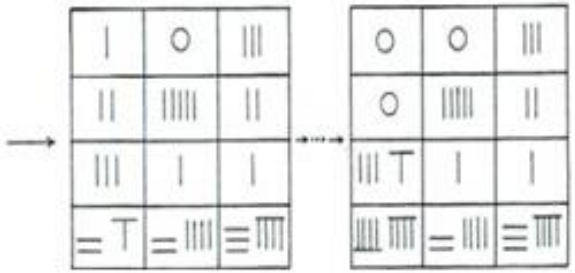
1,2,3,4,5,6,7,8,9



1	2	3
2	3	2
3	1	1
26	34	39



0	0	3
4	5	2
8	1	1
39	24	39



0	0	3
0	5	2
36	1	1
99	24	39

Example 2

Solving the following system of linear equations using the method of Gaussian elimination.

$$x_1 + 2x_2 + 3x_3 + 2x_4 = -1$$
$$-x_1 - 2x_2 - 2x_3 + x_4 = 2$$
$$2x_1 + 4x_2 + 8x_3 + 12x_4 = 4$$

Solution

Example 2

Solving the following system of linear equations using the method of Gaussian elimination.

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Solution

We arrive at the *echelon form* as in the previous example.

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 2 & -1 \\ -1 & -2 & -2 & 1 & 2 \\ 2 & 4 & 8 & 12 & 4 \end{array}\right] \sim, \quad \cdots \sim \left[\begin{array}{cccc|c} 1 & 2 & 3 & 2 & -1 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

Echelon form

This marks the end of the Gaussian elimination of variables from equations.

*Can we simplify the system even more? -> "**Reduced** echelon"*

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 2 & -1 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right] \begin{array}{c} \sim \\ R1+(-2)R3 \\ R2+(-3)R3 \end{array} \left[\begin{array}{cccc|c} 1 & 2 & 3 & 0 & -5 \\ 0 & 0 & 1 & 0 & -5 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

r.e.f.

reduced r.e.f.

$$\sim R1+(-3)R2 \left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 10 \\ 0 & 0 & 1 & 0 & -5 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

行最简形

This matrix is the **reduced echelon form** of the original augmented matrix. The corresponding system of equations is

$$x_1 + 2x_2 = 10$$
$$x_3 = -5$$
$$x_4 = 2$$

Let $x_2 = r$, we get same solution as previously,

$$x_1 = -2r + 10,\ x_2 = r,\ x_3 = -5,\ x_4 = 2$$

Reduced Row Echelon Form

$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{bmatrix}$$

In HW please use ~ or → as standard rule and notations for row equivalent transforms

Definition

行最简形

A matrix is in ***reduced row echelon form*** (rref)

1. If the matrix is in row echelon form.
2. If the first nonzero entry in each row is the only nonzero entry in its column

Each leading 1 is the only nonzero entry in its column.

Examples for Reduced Row Echelon Form

$$\begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 9 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 3 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

r.e.f ✓ r.r.e.f ✓

✗ r.e.f ✗ r.r.e.f

✓ r.r.e.f

✗ r.e.f ✗ r.r.e.f

Q: $[0\ 0\ 1\ 2\ 3]$ r.r.e.f ? ✓

Examples for Reduced Row Echelon Form

$$\begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 9 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 3 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(✓) (×) (✓) (×)

The process of using elementary row operations to transform a matrix into reduced row echelon form is called ***Gauss-Jordan reduction 高斯若尔当消元法***.

The reduced row echelon form of a matrix is **unique**.

Gaussian Elimination → r.e.f / r.r.e.f

Gauss-Jordan reduction → r.r.e.f

Q1: If a matrix is in r.r.e.f, is it in r.e.f ?

✓.

Q2: If a matrix is in S.T.F, is it in r.e.f ?

May not Eg. $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & 4 & 5 & 8 \\ 0 & 0 & 6 & 9 \end{array}\right]$ not r.e.f

$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 8 \end{array}\right]$ ✓ S.T.F & r.e.f

Q3: If a matrix is in r.e.f is it in S.T.F ?

May not. $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 1 \end{array}\right]$ ✓ r.e.f ✗ S.T.F

Q4: A matrix can have many r.e.f. but the r.r.e.f is unique.

Eg. $C = \begin{bmatrix} 1 & 0 & 2 & 10 \\ 0 & 0 & 2 & 20 \end{bmatrix}$

$\xrightarrow{\frac{1}{2}R_2}\begin{bmatrix}1&0&2&1&0\\0&0&1&1&0\end{bmatrix}=A$ is in r.e.f

$\xrightarrow{R_1+R_2}\begin{bmatrix}1&0&3&2&0\\0&0&1&1&0\end{bmatrix}=B$ is in r.e.f

$C \xrightarrow{R_1-R_2} \xrightarrow{\frac{1}{2}R_2}\begin{bmatrix}1&0&0&-1&0\\0&0&1&1&0\end{bmatrix}$ r.r.e.f.

$A \xrightarrow{R_1-2R_2}\begin{bmatrix}1&0&0&-1&0\\0&0&1&1&0\end{bmatrix}$ r.r.e.f

$B \xrightarrow{R_1-3R_2}\begin{bmatrix}1&0&0&-1&0\\0&0&1&1&0\end{bmatrix}$ r.r.e.f.

Q: If a matrix is in S.T.F.

then its r.r.e.f is the identity matrix.

Eg. $I_2=\begin{bmatrix}1&0\\0&1\end{bmatrix}_{2\times 2}$

$I_n=\begin{bmatrix}1&&\\&\ddots&\\&&1\end{bmatrix}_{n\times n}$

S.T.F

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1,n-1} & a_{1n} \\ 0 & a_{22} & \cdots & a_{2,n-1} & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{n-1,n-1} & a_{n-1,n} \\ 0 & 0 & \cdots & 0 & a_{nn} \end{bmatrix}$$

$a_{kk} \neq 0$

$\frac{1}{a_{11}} R_1$

$\frac{1}{a_{22}} R_2$

$\frac{1}{a_{nn}} R_n$

$$\begin{bmatrix} 1 & * & \cdots & * & * \\ 0 & 1 & \cdots & * & * \\ \vdots & \vdots & \ddots & 1 & \vdots \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$

$x_1 = s_1$

$x_2 = s_2$

$\vdots$

$x_n = s_n$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & \cdots & 0 & s_1 \\ 0 & 1 & \cdots & 0 & s_2 \\ \vdots & \vdots & \ddots & 1 & \vdots \\ 0 & 0 & \cdots & 1 & s_n \end{array}\right]$$

I_n

Example

Solving the following system of linear equation

$$x_1 + x_2 + x_3 = 2$$
$$2x_1 + 3x_2 + x_3 = 3$$
$$x_1 - x_2 - 2x_3 = -6$$

Example

Solving the following system of linear equation by Gauss-Jordan reduction

$$x_1 + x_2 + x_3 = 2$$
$$2x_1 + 3x_2 + x_3 = 3$$
$$x_1 - x_2 - 2x_3 = -6$$

Solution

Augmented matrix:

Pivotal row

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ 1 & -1 & -2 & -6 \end{bmatrix}$$

$\sim$ R2+(–2)R1, R3+(–1)R1

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & -2 & -3 & -8 \end{bmatrix}$$

Pivotal row

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & -2 & -3 & -8 \end{array}\right]$$

$\sim$ R1+(–1)R2, R3+(2)R2

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -5 & -10 \end{array}\right]$$

$\sim$ (–1/5)R3

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$\sim$ R1+(–2)R3, R2+R3

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

The solution is

$$x_1 = -1, x_2 = 1, x_3 = 2.$$

(-1, 1, 2) is a solution

Check! Is this a solution of the original system?
Build good habit for doing math problems

$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

Try to turn the matrix into the form below.

$$\begin{bmatrix} & & \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} & & \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} & & \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Step 1: eliminate the first column such that only the leading entry 1 is left
- Step 2: eliminate the second column such that only the leading entry 1 is left in the column
- …

How to determine whether a general linear system is consistent or not ? Very Important

- The augmented matrix and its (reduced) row echelon form are row equivalent matrices. 增广矩阵与其行阶梯形、行最简形矩阵是行等价的。
- The row equivalent matrices share the same solutions 行等价的增广矩阵对应的线性方程组具有相同的解集.
- To determine whether a general linear system is consistent or not, we can determine whether the linear systems corresponding to the (reduced) row echelon form is consistent or not. 判断一个线性方程组是否相容（有解），只需要判断其行最简行对应的线性方程组是否相容（有解））
- The consistency of the linear system corresponding to the (reduced) row echelon form is easy to check.最简阶梯矩阵对应的线性方程组的相容性是很容易判断的（左边系数都是0时，右边是不是0）

Very important!

1. Write the ***augmented matrix*** of the system.
2. Obtain an equivalent augmented matrix in ***reduced echelon form***.
3. Write the system of equations corresponding to the matrix obtained in step 2.

   a) The linear system is inconsistent, if there is one equation reads as $0x_1 + 0x_2 + \cdots + 0x_n = b_n \ (b_n \neq 0)$

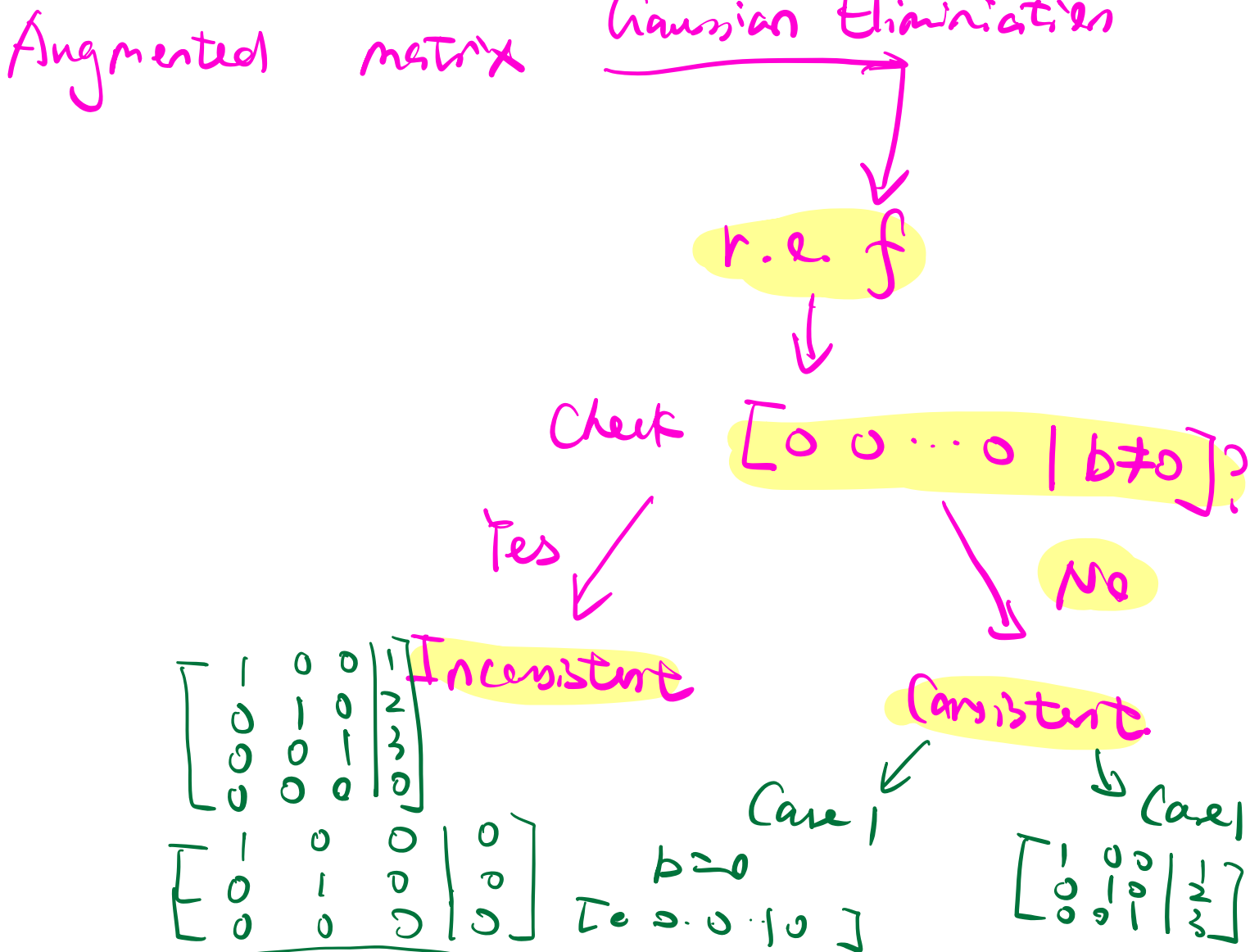
   b) The linear system is consistent:

      i. The linear system has a unique solution if the reduced form is of $[I_n | \ b]$ ***Question: what's the unique solution?***

      ii. The linear system has many solutions if the reduced echelon form has more variables than equations, or if the last row of reduced echelon form consists of all zeros

Ch1: ∀ m×n linear system
1. Consistent or inconsistent?
2. If it is consistent, how many solutions?
3. How to express the solutions?

Q1: To check the consistency of ∀ m×n linear system.



Exercise (very important)

For what values of *h* will the following system be consistent?

$$3x_1 - 9x_2 = 4$$
$$-2x_1 + 6x_2 = h$$

$3x_1 - 9x_2 = 4$

$0 = 0$

$$\left[\begin{array}{cc|c} 3 & -9 & 4 \\ -2 & 6 & h \end{array}\right] \xrightarrow{R_2 + \frac{2}{3}R_1} \left[\begin{array}{cc|c} 3 & -9 & 4 \\ 0 & 0 & h + \frac{8}{3} \end{array}\right]$$

If $h + \frac{8}{3} \neq 0$, $h \neq -\frac{8}{3}$, inconsistent

If $h + \frac{8}{3} = 0$, $h = -\frac{8}{3}$

consistent → infinitely many solutions

Q: $\left[\begin{array}{cc|c} 3 & -9 & 4 \\ -2 & 6+a & h \end{array}\right] \xrightarrow{R_2 + \frac{2}{3}R_1} \left[\begin{array}{cc|c} 3 & -9 & 4 \\ 0 & a & h+\frac{8}{3} \end{array}\right]$

a 可以 $= 0$

For what values of a and h, will the system have

(a) unique solution

(b) infinitely many solutions ?

(c) No solution.

(a) If $a \neq 0$. the 2x2 linear system can be simplified to S.T.F.

the system has the unique solution.

(b) $a=0$ & $h=-\frac{8}{3}$ $\left[\begin{array}{cc|c} 3 & -9 & 4 \\ 0 & 0 & 0 \end{array}\right]$

infinitely many solutions.

(c) $a=0$ & $h+\frac{8}{3} \neq 0$ / $h \neq -\frac{8}{3}$

The linear system will be inconsistent

Exercise (very important)

For what values of *h* will the following system be consistent?

$$\begin{aligned} 3x_1 - 9x_2 &= 4 \\ -2x_1 + 6x_2 &= h \end{aligned}$$

- Reduce to triangular form.

$$\left[\begin{array}{cc|c} 3 & -9 & 4 \\ -2 & 6 & h \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & -3 & \frac{4}{3} \\ -2 & 6 & h \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & -3 & \frac{4}{3} \\ 0 & 0 & h + \frac{8}{3} \end{array}\right]$$

Note: 判断线性方程组的相容性，可以首先求其增广矩阵的（最简）阶梯矩阵

$$0x_1 + 0x_2 = h + \frac{8}{3}$$

- System is consistent only if h=-8/3

Example of inconsistent system

This example illustrates a system that has no solution. Let us try to solve the system

$$x_1 + x_2 + 5x_3 = 3$$
$$x_2 + 3x_3 = -1$$
$$x_1 + 2x_2 + 8x_3 = 3$$

inconsistent.

$$\left[\begin{array}{ccc|c} 1 & 1 & 5 & 3 \\ 0 & 1 & 3 & -1 \\ 1 & 2 & 8 & 3 \end{array}\right] \xrightarrow{R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 5 & 3 \\ 0 & 1 & 3 & -1 \\ 0 & 1 & 3 & 0 \end{array}\right] \xrightarrow{R_3 - R_2}$$

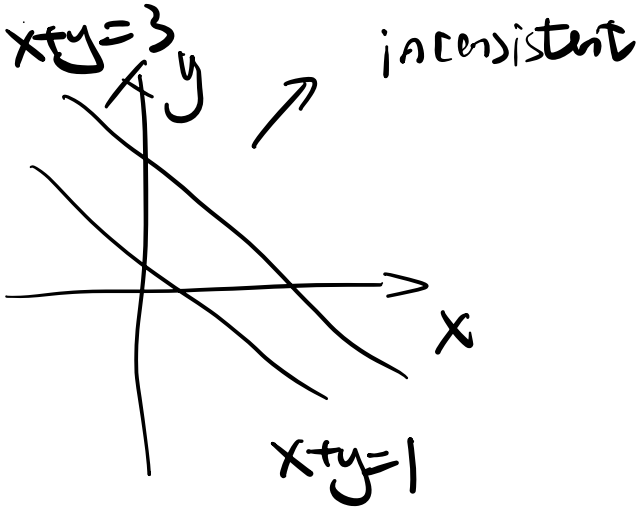
$$\left[\begin{array}{ccc|c} 1 & 1 & 5 & 3 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

$0x_1 + 0x_2 + 0x_3 = 1$

$0 = 1$ Impossible

Inconsistent

Eg. $\begin{cases} x+y=3 \\ x+y=1 \end{cases} \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 1 & 1 & 1 \end{array}\right] \xrightarrow{R_2-R_1}$



$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 0 & -2 \end{array}\right]$

$0 = -2$

Impossible.

Example of inconsistent system

This example illustrates a system that has no solution. Let us try to solve the system

$$x_1 + x_2 + 5x_3 = 3$$
$$x_2 + 3x_3 = -1$$
$$x_1 + 2x_2 + 8x_3 = 3$$

Solution

$$\left[\begin{array}{ccc|c} 1 & 1 & 5 & 3 \\ 0 & 1 & 3 & -1 \\ 1 & 2 & 8 & 3 \end{array}\right] \underset{R3+(-1)R1}{\sim} \left[\begin{array}{ccc|c} 1 & 1 & 5 & 3 \\ 0 & 1 & 3 & -1 \\ 0 & 1 & 3 & 0 \end{array}\right] \underset{\substack{R1+(-1)R2 \\ R3+(-1)R2}}{\sim} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

$$0x_1+0x_2+0x_3=1$$

The system has no solution.

If a linear system is consistent, how to solve it?

How to solve linear systems

by the Gauss-Jordan Elimination高斯若尔当消元法

System of linear equations

1. Write the ***augmented matrix*** of the system.
2. Obtain an equivalent augmented matrix in ***reduced echelon form***. Decide whether the system is consistent. If not, stop; otherwise go to the next step.
3. Write the system of equations corresponding to the matrix obtained in step 2.
4. State the ***solution*** by expressing each lead variable in terms of the free variables and declare the free variables.

Example with unique solution

Solving the following system of linear equation.

$$x_1 - 2x_2 + 4x_3 = 12$$
$$2x_1 - x_2 + 5x_3 = 18$$
$$-x_1 + 3x_2 - 3x_3 = -8$$

Example with unique solution

Solving the following system of linear equation.

$$x_1 - 2x_2 + 4x_3 = 12$$
$$2x_1 - x_2 + 5x_3 = 18$$
$$-x_1 + 3x_2 - 3x_3 = -8$$

Solution

$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & 12 \\ 2 & -1 & 5 & 18 \\ -1 & 3 & -3 & -8 \end{array}\right] \begin{array}{l} R2 + (-2)R1 \\ R3 + R1 \end{array} \sim \left[\begin{array}{ccc|c} 1 & -2 & 4 & 12 \\ 0 & 3 & -3 & -6 \\ 0 & 1 & 1 & 4 \end{array}\right]$$

No free variable ⇓ unique

$$\sim \left(\frac{1}{3}\right)R2 \left[\begin{array}{ccc|c} 1 & -2 & 4 & 12 \\ 0 & 1 & -1 & -2 \\ 0 & 1 & 1 & 4 \end{array}\right] \begin{array}{l} R1 + (2)R2 \\ R3 + (-1)R2 \end{array} \sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 8 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 2 & 6 \end{array}\right]$$

$$\sim \left(\frac{1}{2}\right)R3 \left[\begin{array}{ccc|c} 1 & 0 & 2 & 8 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 3 \end{array}\right] \begin{array}{l} R1 + (-2)R3 \\ R2 + R3 \end{array} \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

x_1 x_2 x_3

(2, 1, 3)

x_1, x_2, x_3 are leading variables

Solution: $x_1 = 2, x_2 = 1, x_3 = 3$

Example with many solutions

Solve, if possible, the system of equations

$$3x_1 - 3x_2 + 3x_3 = 9$$
$$2x_1 - x_2 + 4x_3 = 7$$
$$3x_1 - 5x_2 - x_3 = 7$$

Example with many solutions

Solve, if possible, the system of equations

$$3x_1 - 3x_2 + 3x_3 = 9$$
$$2x_1 - x_2 + 4x_3 = 7$$
$$3x_1 - 5x_2 - x_3 = 7$$

If doesn't have [0 0 ... 0 | b≠0]

Solution

$$\left[\begin{array}{ccc|c} 3 & -3 & 3 & 9 \\ 2 & -1 & 4 & 7 \\ 3 & -5 & -1 & 7 \end{array}\right] \xrightarrow{\left(\frac{1}{3}\right)R1} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 2 & -1 & 4 & 7 \\ 3 & -5 & -1 & 7 \end{array}\right] \xrightarrow[R3+(-3)R1]{R2+(-2)R1} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & -2 & -4 & -2 \end{array}\right]$$

Consistent → infinitely

has a free variable

$$\xrightarrow[R3+2R2]{R1+R2} \left[\begin{array}{ccc|c} 1 & 0 & 3 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{array}{l} x_1 + 3x_3 = 4 \\ x_2 + 2x_3 = 1 \end{array} \Rightarrow \begin{array}{l} x_1 = -3x_3 + 4 \\ x_2 = -2x_3 + 1 \end{array}$$

x_1 x_2 x_3

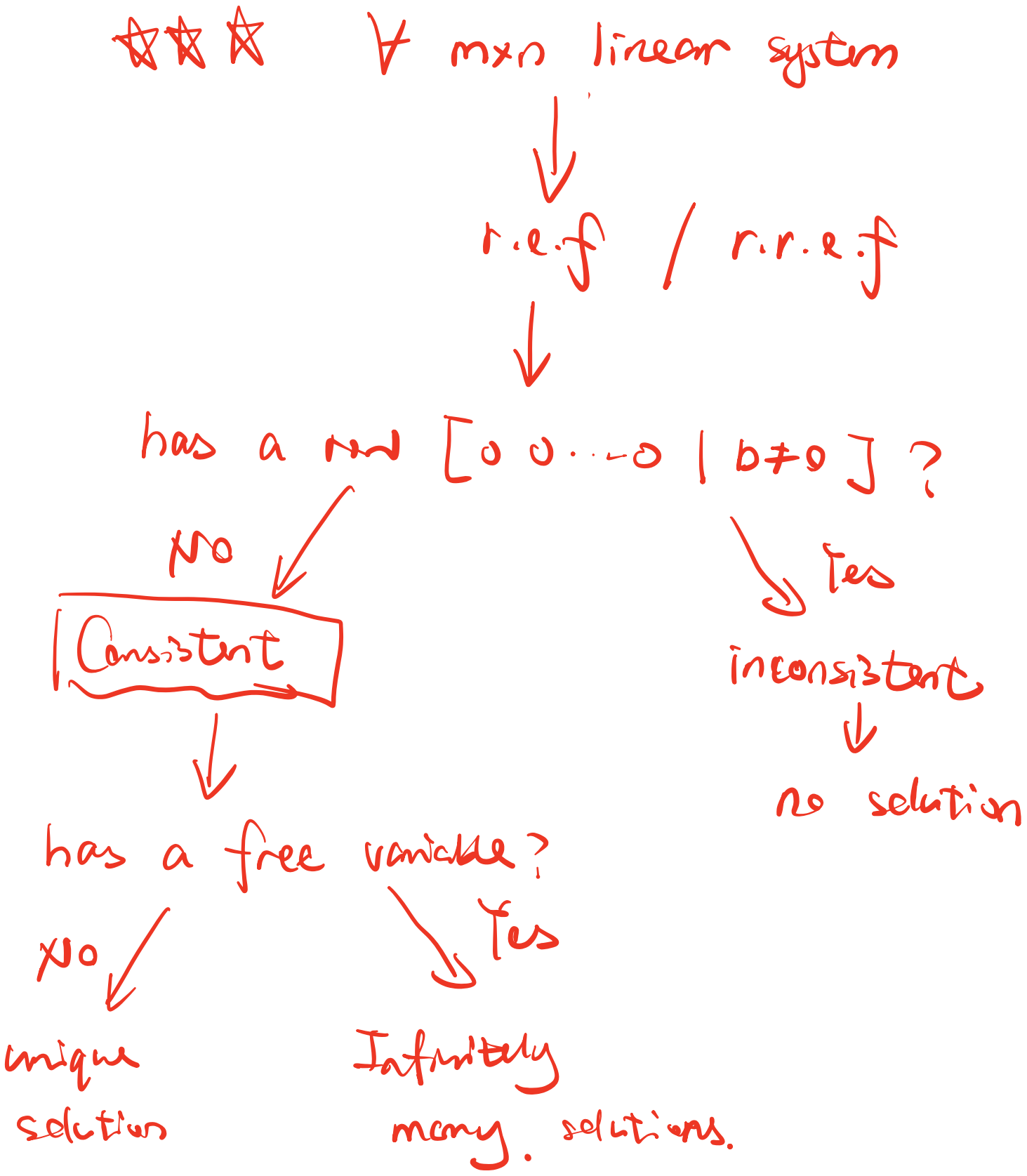
The general solution to the system is

x_1, x_2 are leading variables.

x_3 is a free variable

$$x_1 = -3r + 4$$
$$x_2 = -2r + 1$$
$$x_3 = r \quad , \text{where } r \text{ is real number (called a parameter).}$$

$(-3r+4, -2r+1, r)$, $\forall r \in \mathbb{R}$ is the solution


∀ mxn linear system
r.e.f / r.r.e.f
has a row [0 0 ... 0 | b≠0] ?
No
Yes
Consistent
inconsistent
no solution
has a free variable?
No
Yes
unique solution
Infinitely many solutions.

Eg. $\left[\begin{array}{ccccc|c} 1 & 1 & 2 & 3 & 5 & 1 \\ 0 & 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$

3×5

$3a+5$

How to determine free variables and leading variables 如何确定自由变量

$$\xrightarrow[\substack{R1+R2\\R3+R2}]{}\left[\begin{array}{ccc|c} 1 & 0 & 3 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

x_1 x_2 x_3 **b**

Free variable $x_3 = r$

Leading variable $x_1 = 4 - 3r$

Leading variable $x_2 = 1 - 2r$

- Unknowns corresponding to columns with leading entries are leading variables
- Unknowns corresponding to columns with no leading entries are free variables. Free variables can be any number. 自由变量可以是任意值，对应于没有leading entries的列。

$\forall$ $m \times n$ linear system

m = number of equations

n = number of variables

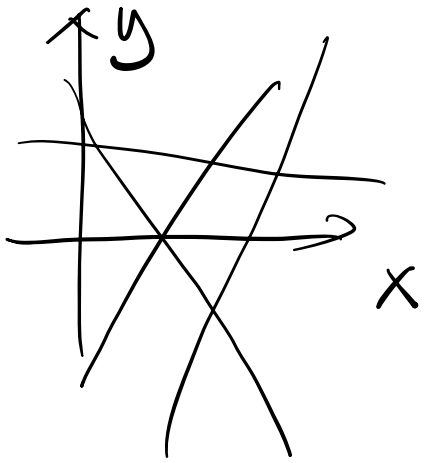
$\begin{cases} ① \ m > n \\ ② \ m = n \\ ③ \ m < n \end{cases}$

under-determined

① $m > n$

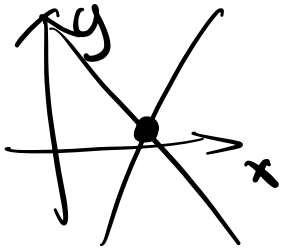
over-determined

Eg. 4×2



② $m = n$

2×2



③ $m < n$

$x + 2y + z = 3$

$x \quad -3z = 1$

Ex: under/over determined system

Solve the system of equations

$$x_1 + 2x_2 - x_3 + 3x_4 = 4$$
$$2x_1 + 4x_2 - 2x_3 + 7x_4 = 10$$
$$-x_1 - 2x_2 + x_3 - 4x_4 = -6$$

3×4

$m < n$

under-determined

Ex: under/over determined system

Solve the system of equations

$$x_1+2x_2-x_3+3x_4=4$$
$$2x_1+4x_2-2x_3+7x_4=10$$
$$-x_1-2x_2+x_3-4x_4=-6$$

亚定的

***Underdetermined system*:**
The number of variables
> The number of equations.
⇒ 'usually' many sol.

Solution

$$\begin{bmatrix} 1 & 2 & -1 & 3 & 4 \\ 2 & 4 & -2 & 7 & 10 \\ -1 & -2 & 1 & -4 & -6 \end{bmatrix} \xrightarrow[\text{R3+R1}]{\text{R2+(-2)R1}} \begin{bmatrix} 1 & 2 & -1 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 & -2 \end{bmatrix}$$

$$\xrightarrow[\text{R3+R2}]{\text{R1+(-3)R2}} \begin{bmatrix} 1 & 2 & -1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1+2x_2-x_3=-2$

$x_4=2$

x_1 x_2 x_3 x_4

超定的

***Overdetermined system*:**
The number of variables
< The number of equations.
⇒ 'usually' inconsistent

Thus, from row 1: $x_1+2x_2-x_3=-2$

-> $x_1=-2-2x_2+x_3$;

from row 2: $x_4=2$.

The solution is $(-2r+s-2, r, s, 2)$ for $r, s \in R$

x_1, x_4 are leading variables.
x_2, x_3 are
free variables

let $x_2 = \alpha$, $x_3 = \beta$. $\forall \alpha, \beta \in \mathbb{R}$

$$x_1 = -2 - 2x_2 + x_3 = -2 - 2\alpha + \beta$$

$$(-2 - 2\alpha + \beta, \alpha, \beta, 4) \quad \forall \alpha, \beta \in \mathbb{R}.$$

How to determine free variables and leading variables 如何确定自由变量

$$\xrightarrow[R3+R2]{R1+(-3)R2} \left[\begin{array}{cccc|c} \textcircled{1} & 2 & -1 & 0 & -2 \\ 0 & 0 & 0 & \textcircled{1} & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$x_1 \quad x_2 \quad x_3 \quad x_4$$

$$r \quad s$$

Parallel Computing with Multiple right-hand sides: how to solve fast

- If two linear systems share the same coefficient matrix but with different right hand sides, say,

$A\boldsymbol{x} = \boldsymbol{b}_1$ and $A\boldsymbol{x} = \boldsymbol{b}_2$

- We do not need to solve the two linear systems twice.
- We can solve the two linear systems at the same time (and reduce almost half computations), by writing the argumented matrix together

$$[A \,|\boldsymbol{b}_1 \; \boldsymbol{b}_2]$$

Coefficient matrix

RSH1

RSH2

Example: multiple right hand sides

Solving the following three systems of linear equation, all of which have the same matrix of coefficients.

$$\begin{aligned} x_1 - x_2 + 3x_3 &= b_1 \\ 2x_1 - x_2 + 4x_3 &= b_2 \\ -x_1 + 2x_2 - 4x_3 &= b_3 \end{aligned} \quad \text{for} \quad \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 11 \\ -11 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix} \text{ in turn}$$

Solution

Example: multiple right hand sides

Solving the following three systems of linear equation, all of which have the same matrix of coefficients.

$$\begin{aligned} x_1 - x_2 + 3x_3 &= b_1 \\ 2x_1 - x_2 + 4x_3 &= b_2 \\ -x_1 + 2x_2 - 4x_3 &= b_3 \end{aligned} \quad \text{for} \quad \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 11 \\ -11 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix} \text{ in turn}$$

Solution

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 3 & 8 & 0 & 3 \\ 2 & -1 & 4 & 11 & 1 & 3 \\ -1 & 2 & -4 & -11 & 2 & -4 \end{array}\right] \underset{\substack{\text{R2+(–2)R1} \\ \text{R3+R1}}}{\sim} \left[\begin{array}{ccc|ccc} 1 & -1 & 3 & 8 & 0 & 3 \\ 0 & 1 & -2 & -5 & 1 & -3 \\ 0 & 1 & -1 & -3 & 2 & -1 \end{array}\right]$$

$$\underset{\substack{\text{R1+R2} \\ \text{R3+(–1)R2}}}{\sim} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & -5 & 1 & -3 \\ 0 & 0 & 1 & 2 & 1 & 2 \end{array}\right] \underset{\substack{\text{R1+(–1)R3} \\ \text{R2+2R3}}}{\sim} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & -2 \\ 0 & 1 & 0 & -1 & 3 & 1 \\ 0 & 0 & 1 & 2 & 1 & 2 \end{array}\right]$$

The solutions to the three systems are

$$\begin{cases} x_1 = 1 \\ x_2 = -1, \\ x_3 = 2 \end{cases} \begin{cases} x_1 = 0 \\ x_2 = 3, \\ x_3 = 1 \end{cases} \begin{cases} x_1 = -2 \\ x_2 = 1 \\ x_3 = 2 \end{cases}.$$

Homogeneous System of linear Equations

Definition

A system of linear equations is said to be ***homogeneous*** (齐次的)if all the constant terms on the right hand side are all zeros.

The right hand side ***b***

Theorem

A system of homogeneous linear equations in *n* variables always has the solution $x_1 = 0, x_2 = 0, \ldots, x_n = 0$. This solution is called the ***trivial solution***. 齐次线性方程组均有平凡解

Example

Observe that $x_1 = 0, x_2 = 0, x_3 = 0$ is a solution for the linear system

$$x_1 + 2x_2 - 5x_3 = 0$$
$$-2x_1 - 3x_2 + 6x_3 = 0$$

Homogeneous System of linear Equations

In addition to the trivial solution, there might be other solutions.

$$\begin{cases} x_1 + 2x_2 - 5x_3 = 0 \\ -2x_1 - 3x_2 + 6x_3 = 0 \end{cases}$$

The system has other nontrivial solutions.

$$\left[\begin{array}{ccc|c} 1 & 2 & -5 & 0 \\ -2 & -3 & 6 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 0 & 1 & -4 & 0 \end{array}\right]$$

$$\therefore x_1 = -3r,\ x_2 = 4r,\ x_3 = r$$

Theorem

A system of homogeneous linear equations that has more variables than equations has many solutions. 齐次线性方程组的未知量大于方程的个数时具有(无穷) 多个解.

Exercise

Is this system consistent?

$$-x_2 - x_3 + x_4 = 0$$
$$x_1 + x_2 + x_3 + x_4 = 6$$
$$2x_1 + 4x_2 + x_3 - 2x_4 = -1$$
$$3x_1 + x_2 - 2x_3 + 2x_4 = 3$$

Exercise

Is this system consistent?

$$-x_2 - x_3 + x_4 = 0$$
$$x_1 + x_2 + x_3 + x_4 = 6$$
$$2x_1 + 4x_2 + x_3 - 2x_4 = -1$$
$$3x_1 + x_2 - 2x_3 + 2x_4 = 3$$

This system can be reduced to a **strictly triangular system**:

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & -3 & -2 & -13 \\ 0 & 0 & 0 & -1 & -2 \end{array}\right]$$

This is sufficient to see that the system is consistent and unique. Why?

Exercise

Is this system consistent?

$$\begin{aligned} 3x_2 - 6x_3 &= 8 \\ x_1 - 2x_2 + 3x_3 &= -1 \\ 5x_1 - 7x_2 + 9x_3 &= 0 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 0 & 3 & -6 & 8 \\ 1 & -2 & 3 & -1 \\ 5 & -7 & 9 & 0 \end{array}\right]$$

Exercise

Is this system consistent?

$$\begin{aligned} 3x_2 - 6x_3 &= 8 \\ x_1 - 2x_2 + 3x_3 &= -1 \\ 5x_1 - 7x_2 + 9x_3 &= 0 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 0 & 3 & -6 & 8 \\ 1 & -2 & 3 & -1 \\ 5 & -7 & 9 & 0 \end{array}\right]$$

- Row operations produce:

$$\left[\begin{array}{ccc|c} 0 & 3 & -6 & 8 \\ 1 & -2 & 3 & -1 \\ 5 & -7 & 9 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & 3 & -6 & 8 \\ 0 & 3 & -6 & 5 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & 3 & -6 & 8 \\ 0 & 0 & 0 & -3 \end{array}\right]$$

- Equation notation of triangular form:

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= -1 \\ 3x_2 - 6x_3 &= 8 \\ 0x_3 &= -3 \end{aligned}$$

Never true!
The original system is inconsistent!

Learning outcomes

1. Be able to tell what is row echelon form and reduced row echelon form (能够辨识什么是梯形形式和简化梯形形式)
2. Be able to obtain the (reduced) Row Echelon Form of a matrix 会将一个矩阵通过行初等变换转化为等价的（简化)梯形形式
3. Be able to determine whether a linear system is consistent according to its row echelon form. 能够通过线性方正组增广矩阵的等价梯形形式判断方程组是否有解， 有几个解答
4. Be able to obtain the solutions of consist linear system, 会求解相容线性方程组的解。
5. Be able to solve under determined linear systems, determine the free variables according to the row echelon form. 能够求解欠定方程组，并根据梯形形式确定自由变量。